Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 January 2026
Team ID	Individual
Project Name	EV Market Performance & Sustainability Analytics
Maximum Marks	4 Marks

Data Flow Diagrams:

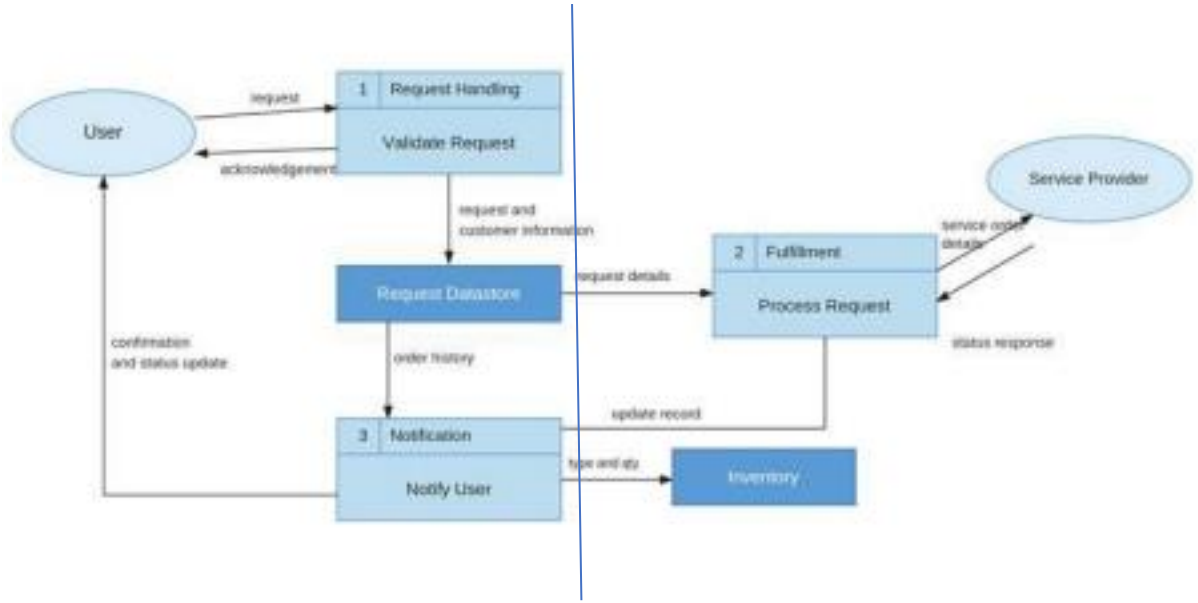
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).

Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account / dashboard	High	Sprint-1
		USN-2	As a user, I will receive confirmation email once I have registered for the application	I can receive confirmation email & click confirm	High	Sprint-1
		USN-3	As a user, I can register for the application through Facebook	I can register & access the dashboard with Facebook Login	Low	Sprint-2
		USN-4	As a user, I can register for the application through Gmail		Medium	Sprint-1
	Login	USN-5	As a user, I can log into the application by entering email & password		High	Sprint-1
	Dashboard					
Customer (Web user)						
Customer Care Executive						
Administrator						

EV ANALYTICS — DATA FLOW DIAGRAM (LEVEL 0)

Data source → Tableau data preparation and analysis → Interactive dashboards → Business users

EV Dataset **(CSV / Excel)**	**1. Data Preparation** **Check types, nulls, fields**	**Prepared EV Data**
2. Tableau Analysis **KPIs, sales, range, battery**	**3. Dashboard Views** **Charts, filters, tooltips**	**Business Users** **Strategy / BI / Research**
Input: EV records	Output: Visual insights	Decision: Explore / compare

DFD explanation: The CSV/Excel dataset is connected to Tableau. Fields are checked and prepared, measures are aggregated into KPIs and analytical views, and the dashboard presents the results to users for comparison and decision support.

COMPLETED USER STORIES

User Type	Functional Requirement (Epic)	Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Market Strategy Director	Sales Analysis	US-01	As a strategy director, I want to compare annual EV sales by year and category so I can identify important segments.	Sales trend updates by model year and category; labels and tooltips are readable.	High	Sprint 1
Business Intelligence Manager	Country Analysis	US-02	As a BI manager, I want to compare sales by manufacturing country so I can explore regional differences.	Country bars show sales values and can be compared consistently.	High	Sprint 1
Market Analyst	Category Analysis	US-03	As a market analyst, I want to view sales by vehicle category so I can compare segment contribution.	Category chart displays category names and sales values.	High	Sprint 1
Battery Researcher	Battery Technology	US-04	As a researcher, I want to compare sales share by battery type to understand technology distribution.	Battery chart displays LFP, NCA, NMC, Sodium-Ion, and Solid-State shares where available.	High	Sprint 2
Product Manager	Driving Range	US-05	As a product manager, I want to compare driving range across categories to identify variation.	Range view shows category-wise values with useful tooltips.	Medium	Sprint 2
Sustainability Analyst	CO2 Savings	US-06	As a sustainability analyst, I want to review annual CO2 savings to assess the recorded environmental metric.	CO2 KPI is visible with unit and calculation definition.	High	Sprint 2
Executive User	KPI Overview	US-07	As an executive, I want key metrics displayed at the top so I can understand overall performance quickly.	Four KPI cards show total sales, average price, average range, and annual CO2 saved.	High	Sprint 1
Dashboard User	Interactive Filters	US-08	As a dashboard user, I want to filter by model year, category, and battery type to focus on selected data.	Selected filters affect the intended worksheets and can be reset.	High	Sprint 3
Manufacturer Analyst	Manufacturer Performance	US-09	As an analyst, I want to explore manufacturer performance and customer ratings.	Manufacturer chart exposes relevant measures; rating scale and aggregation are validated.	Medium	Sprint 3
Research Analyst	Charging & Efficiency	US-10	As a research analyst, I want to explore charging power against energy efficiency.	Scatter plot uses verified units, aggregation, and readable axes.	Medium	Sprint 3
Project Owner	Reporting	US-11	As a project owner, I want to publish or export the dashboard for stakeholders.	Dashboard can be saved or published using an approved Tableau destination.	Medium	Sprint 4
Data Analyst	Data Quality	US-12	As a data analyst, I want to validate fields and calculations so that the dashboard is trustworthy.	Field types, nulls, duplicates where relevant, and KPI calculations are checked and documented.	High	Sprint 1